



Department of Electrical and Electronic Engineering

EEE499P :Final Year Design Project (FYDP)

Weekly Update

Week Number: 03

Date: From 26 Oct 2025 To 02Nov 2025.

ATC Panel:

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Week 3 Update: Functional Analysis.

Initial Problem Statement:

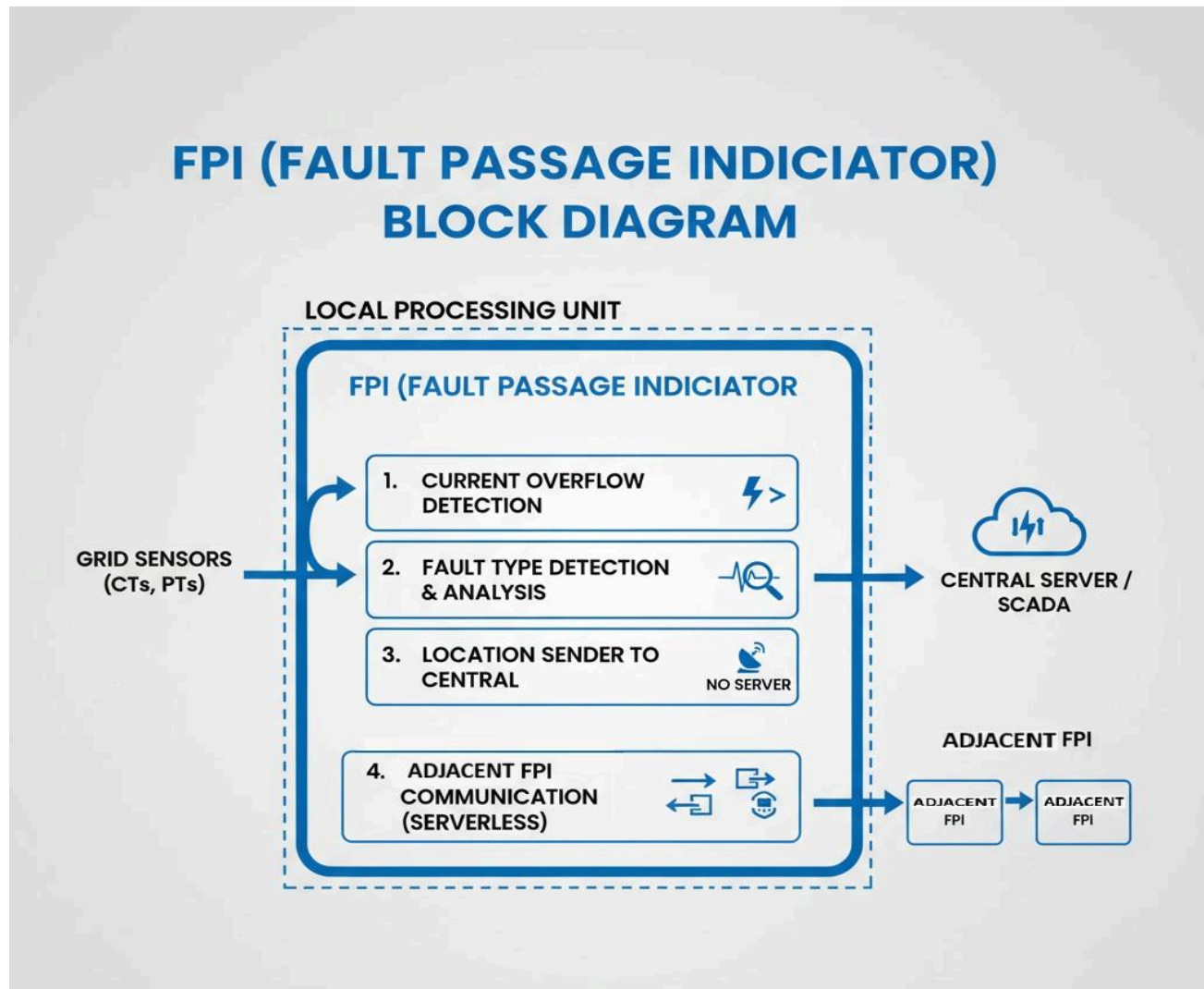
In power distribution networks, faults such as phase, earth, and line-to-line faults frequently occur due to system erosion, weather conditions and load fluctuations. Traditional fault detection systems rely on manual inspection and delayed response, resulting in longer use of time, reduced reliability and higher maintenance costs. Therefore, for utility operators, there is a need to develop a device that can accurately detect, classify and locate faults in real time, ensuring an improved system efficiency.

Proposed Solution: Fault Passage Indicator.

Parameter:

- 1.Current OverFlow
- 2.Position Detection
- 3.Fault types(Phase Fault, Earth Fault,Line to Line fault)
- 4.Communication

Block Diagram:



Sub Function(Verb and Noun):

1. Detects Current Overflow.
2. Detects and analyzes the type of fault.
3. Sends Location to the central server.
4. Communicates with adjacent FPI without any local server.

Sub Systems:

1. Processing Unit:

Will consist of a microcontroller(RL78/I1C) or an arduino/esp32 to process the sensor signals. Identify the fault type and all real time current flow through each line and send the signal to the central server.

2. Communication Unit:

Collects real-time status and fault data from field devices through optical fiber, RF, or cellular networks, transmits it to the central SCADA/ADMS for processing and automated decision-making, and then gives control commands or alerts back to field equipment and maintenance crews to enable rapid fault isolation and service restoration.

3. Current Sensor Unit:

The current sensor unit measures AC current (2A–5000A) of the transmission line through Hall effect/ Rogowski Coil and sends a signal to the processing unit. Divided into 0–500A and 500–1000A ranges, it minimizes noise and prevents saturation. The analog output is converted by a 24-bit delta-sigma ADC in the RL78/I1C microcontroller for precise, real-time fault detection.

4. Power Unit:

The Power Unit ensures stable operation of the system using the RL78/I1C microcontroller for control, EEPROM (AT24C512) for data storage, and a CR2032 coin cell to power the Real-Time Clock for accurate timestamps during outages. It is low-power, reliable, and maintains system data after interruptions.

5. Base Station Unit:

The base station unit receives the data from the FPI and identifies the location of the fault FPI with Geographic Information System and also shows the current flow in real time through all the FPI and alerts the user about any fault if taken place.

6. Indicator Unit:

The Indicator Unit uses LEDs to display system status, showing power ON/OFF, normal operation, faults, or power failure. When a current spike above 500A or power loss occurs, LEDs respond within milliseconds, enabling quick fault detection and improving system reliability.